

Positivities and Cones in the Kähler MMP

Kähler MMP

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The aim of this note is to give a brief introduction to positivities and cones in the Kähler minimal model program. On a general compact Kähler variety there are no divisors to work with, so that the positivity notion has to be reformulated on *cohomology classes and the curvature current of singular metrics*. This note is organized as follows: Section 1 introduce the working space, Bott–Chern cohomology $H_{BC}^{1,1}(X)$, and the numerical spaces $N^1(X)$, $N_1(X)$. We then introduce several positivity notions include: Kählerness, bigness, pseudo-effectiveness, nefness and semi-ampleness. We end this note by introducing Mori cone and generalized Mori cone with some dualities of these cones.

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The major references are [DP04], [Bou04], [HP16], [DH24], [DHP24], [DHY23], [BDPP13] and [Dem12, Ch. 18–19].

1 Setting: Bott–Chern cohomology and the numerical spaces

Convention 1.1. Throughout, X is a normal compact complex variety in Fujiki’s class $\mathcal{C}$ unless stated otherwise, and $n = \dim X$. All sheaves below are sheaves of *real-valued* objects. We write $\mathcal{A}_X^0$ for smooth functions $U \rightarrow \mathbb{R}$ (on a singular X : restrictions of smooth functions under a local embedding $U \hookrightarrow \Omega \subset \mathbb{C}^N$), $\mathcal{D}_X$ for degree-0 currents, and $\mathcal{H}_X$ for the sheaf of **pluriharmonic** functions, locally $u = \operatorname{Im} f$ with f holomorphic. The operator d^c is normalised so that $dd^c = \frac{i}{\pi} \partial \bar{\partial}$; the constant plays no role below.

Observation 1.2. $\mathcal{H}_X = \ker(dd^c)$.

For $\mathcal{A}_X^0$ this is the definition of pluriharmonicity. For $\mathcal{D}_X$ it is a genuine theorem: a *distribution* u with $dd^c u = 0$ satisfies $\Delta u = 0$, so by Weyl’s lemma (hypoellipticity of the Laplacian) it is automatically a smooth pluriharmonic function. This is what makes the form-version and the current-version of everything below agree.

1.1 Bott–Chern cohomology

Definition 1.3 (Höring–Peternell, [HP16, Definition 3.1]). Let X be an irreducible reduced complex space. The **Bott–Chern cohomology** of X in bidegree $(1, 1)$ is

$$H_{\text{BC}}^{1,1}(X) := H^1(X, \mathcal{H}_X).$$

On a complex manifold this agrees with the two familiar descriptions,

$$H_{\text{BC}}^{p,q}(X, \mathbb{C}) = \left(\frac{\ker \partial \cap \ker \bar{\partial}}{\operatorname{im} \partial \bar{\partial}} \right)^{p,q} = \left(\frac{\ker d}{\operatorname{im} \partial \bar{\partial}} \right)^{p,q},$$

and with the same quotient formed using closed $(1, 1)$ -currents: every closed $(1, 1)$ -current is Bott–Chern cohomologous to a smooth form. It is the sheaf-theoretic description of

Definition 1.3 that survives on a singular space, which is why it is taken as the definition here.

Definition 1.4. A $(1, 1)$ -form (resp. current) with local potentials on X is a global section of $\mathcal{A}_X^0/\mathcal{H}_X$ (resp. $\mathcal{D}_X/\mathcal{H}_X$).

Unwinding the sheaf quotient, such a section is exactly the data of an open cover $\{U_i\}$, functions $\varphi_i \in \mathcal{A}^0(U_i)$ — the **local potentials** — and the compatibility $\varphi_i - \varphi_j \in \mathcal{H}_X(U_i \cap U_j)$ on overlaps. By **Observation 1.2** the compatibility gives $dd^c \varphi_i = dd^c \varphi_j$ on overlaps, so the local forms glue to one global closed real $(1, 1)$ -form θ with $\theta|_{U_i} = dd^c \varphi_i$. No global potential is asserted to exist.

Proposition 1.5. Every Bott–Chern class is represented by a $(1, 1)$ -form (equivalently, a current) with local potentials.

Proof. The sheaf $\mathcal{A}_X^0$ is fine, hence acyclic, so $H^1(X, \mathcal{A}_X^0) = 0$. The connecting map of

$$0 \rightarrow \mathcal{H}_X \rightarrow \mathcal{A}_X^0 \rightarrow \mathcal{A}_X^0/\mathcal{H}_X \rightarrow 0$$

therefore gives a surjection $H^0(X, \mathcal{A}_X^0/\mathcal{H}_X) \twoheadrightarrow H^1(X, \mathcal{H}_X) = H_{\text{BC}}^{1,1}(X)$. ■

Remark 1.6. The local potential, not the class, is what the theory actually reads. Positivity, Lelong numbers and the whole singularity theory of Sections 2.2 and 2.3 are conditions on φ_i . On a singular X there is no other way to define any of them.

1.2 Comparison with $H^2(X, \mathbb{R})$: where rational singularities enter

Proposition 1.7. Let X be a normal compact complex space in class $\mathcal{C}$ with *rational singularities*. Then the natural map

$$H_{\text{BC}}^{1,1}(X) \hookrightarrow H^2(X, \mathbb{R})$$

is injective. In particular a Bott–Chern class may be treated as a topological class, and $H_{\text{BC}}^{1,1}(X)$ is finite-dimensional.

Proof. The sheaf $\mathcal{H}_X$ sits in the fundamental exact sequence

$$0 \rightarrow \mathbb{R} \rightarrow \mathcal{O}_X \xrightarrow{2\text{Im}} \mathcal{H}_X \rightarrow 0,$$

(a holomorphic function with vanishing imaginary part is constant, by the Cauchy–Riemann equations) whose long exact sequence reads

$$H^1(X, \mathbb{R}) \rightarrow H^1(X, \mathcal{O}_X) \rightarrow H_{\text{BC}}^{1,1}(X) \rightarrow H^2(X, \mathbb{R}) \rightarrow H^2(X, \mathcal{O}_X).$$

So $H_{\text{BC}}^{1,1}(X) \rightarrow H^2(X, \mathbb{R})$ is injective precisely when $H^1(X, \mathbb{R}) \rightarrow H^1(X, \mathcal{O}_X)$ is surjective. We check the latter on a resolution $\pi: \hat{X} \rightarrow X$, using two comparison isomorphisms. In each, normality gives injectivity for free and rationality is what gives surjectivity.

Coherent side. Normality gives $\pi_* \mathcal{O}_{\hat{X}} = \mathcal{O}_X$, and rational singularities are by definition $R^q \pi_* \mathcal{O}_{\hat{X}} = 0$ for $q > 0$. So the Leray spectral sequence degenerates and $H^k(X, \mathcal{O}_X) \cong H^k(\hat{X}, \mathcal{O}_{\hat{X}})$ for all k .

Topological side. Here π has connected fibres by Zariski’s main theorem, so $\pi_* \mathbb{R}_{\hat{X}} = \mathbb{R}_X$, and the five-term exact sequence of Leray reads

$$0 \rightarrow H^1(X, \mathbb{R}) \rightarrow H^1(\hat{X}, \mathbb{R}) \rightarrow H^0(X, R^1 \pi_* \mathbb{R}) \rightarrow H^2(X, \mathbb{R}).$$

Injectivity is automatic on any normal X ; what is needed is $R^1 \pi_* \mathbb{R} = 0$, and this is exactly where rationality is spent.

In the commutative square formed by these two isomorphisms and the two maps $H^1(-, \mathbb{R}) \rightarrow H^1(-, \mathcal{O})$, surjectivity on X is equivalent to surjectivity on $\hat{X}$. There, degree-one Hodge decomposition $H^1(\hat{X}, \mathbb{C}) = H^{1,0} \oplus H^{0,1}$ makes $H^1(\hat{X}, \mathbb{R}) \rightarrow H^1(\hat{X}, \mathcal{O}_{\hat{X}})$ an isomorphism of *real* vector spaces. ■

Remark 1.8. This is the first of many places where **rational singularities** does real work, and it is what one is actually buying: a well-defined intersection theory on $H_{\text{BC}}^{1,1}(X)$, inherited from $H^2(X, \mathbb{R})$.

1.3 Functoriality of Bott–Chern classes

A current cannot in general be pulled back. A $(1, 1)$ -current *with local potentials* can: one pulls back the potentials, and the pulled-back object again has local potentials. So f^* is defined on $H_{\text{BC}}^{1,1}$. What is not automatic is injectivity.

Theorem 1.9 ([HP16, Lemma 3.3]; [DHP24, Lemma 8.7]). *Let $f: X \rightarrow Y$ be a proper morphism with connected fibres between normal compact complex spaces with **rational singularities**. Assume either*

1. $X, Y \in \mathcal{C}$ and f is bimeromorphic, or
2. *there is an effective $\mathbb{Q}$ -divisor $B \geq 0$ with (X, B) klt, $-(K_X + B)$ f -nef and f -big, and f projective.*

Then

$$f^*: H_{\text{BC}}^{1,1}(Y) \rightarrow H_{\text{BC}}^{1,1}(X), \quad f^*: H^2(Y, \mathbb{R}) \rightarrow H^2(X, \mathbb{R})$$

are both injective, and

$$\text{Im}(f^*) = \{\alpha \in H_{\text{BC}}^{1,1}(X) : \alpha \cdot C = 0 \text{ for every curve } C \subset X \text{ contracted by } f\}.$$

Moreover such an α equals $f^[\omega_Y]$ for a smooth real closed $(1, 1)$ -form ω_Y with local potentials on Y .*

Proof of injectivity. Apply the five-term exact sequence of the Leray spectral sequence to $\mathcal{F} = \mathcal{H}_X$, using $f_*\mathcal{H}_X = \mathcal{H}_Y$ (the direct image of the sheaf of pluriharmonic functions under a proper bimeromorphic morphism is again the sheaf of pluriharmonic functions):

$$0 \rightarrow H^1(Y, \mathcal{H}_Y) \rightarrow H^1(X, \mathcal{H}_X) \rightarrow H^0(Y, R^1 f_* \mathcal{H}_X) \rightarrow H^2(Y, \mathcal{H}_Y) \rightarrow H^2(X, \mathcal{H}_X).$$

Exactness at the second place is injectivity of $f^*: H_{\text{BC}}^{1,1}(Y) \rightarrow H_{\text{BC}}^{1,1}(X)$. For the topological statement, the hypotheses give $R^q f_* \mathcal{O}_X = 0$ for $q \geq 1$ and $R^1 f_* \mathbb{R} = 0$, so Leray for $\mathbb{R}$ gives $0 \rightarrow H^2(Y, \mathbb{R}) \rightarrow H^2(X, \mathbb{R}) \rightarrow H^0(Y, R^2 f_* \mathbb{R})$. ■

💡 **Idea for the description of $\text{Im}(f^*)$.** Pushing the sequence $0 \rightarrow \mathbb{R} \rightarrow \mathcal{O}_X \rightarrow \mathcal{H}_X \rightarrow 0$ forward and using $R^{\geq 1} f_* \mathcal{O}_X = 0$, $R^1 f_* \mathbb{R} = 0$ gives an isomorphism $R^1 f_* \mathcal{H}_X \cong R^2 f_* \mathbb{R}$. This makes the two Leray sequences above into a commutative ladder with vertical injections from Proposition 1.7 and an isomorphism on the right. A class dies in $H^0(Y, R^2 f_* \mathbb{R})$ exactly when it is orthogonal to every 2-cycle contained in a fibre; since H_2 of a fibre is generated by algebraic cycles, that is exactly the stated numerical condition. Full details are in [HP16, DHP24].

Remark 1.10. This theorem is the transcendental half of the projective implication “ $L \cdot R = 0 \Rightarrow L = f^* L_Z$ ”. The descent statement is available cheaply here; what is hard is the *second* step, turning the descended class into a Kähler class downstairs. That is the content of the base point free theorem (Theorem 2.52).

Push-forward is more delicate: currents always push forward, but on a singular space the push-forward of a current with local potentials need not have local potentials.

1.4 The numerical spaces $N^1(X)$ and $N_1(X)$

Definition 1.11 ([HP16, Definitions 3.6 and 3.8]). Let X be a normal compact complex space in class $\mathcal{C}$.

1. $N^1(X) := H_{\text{BC}}^{1,1}(X)$.
2. $N_1(X)$ is the vector space of real closed currents of bidimension $(1, 1)$ modulo the relation $T_1 \equiv T_2 \iff T_1(\eta) = T_2(\eta)$ for every real closed $(1, 1)$ -form η with local potentials.

Remark 1.12. Note the deliberate choice in (1): in the analytic setting one does *not* quotient $H_{\text{BC}}^{1,1}$ further by numerical equivalence. The transcendental N^1 is the Bott–Chern group. The algebraic N^1 generated by numerical classes of irreducible divisors is in general far too small to carry the theory.

Proposition 1.13 ([HP16, Proposition 3.9]). *Let X be a normal compact complex space of dimension n with **rational singularities**, in class $\mathcal{C}$. Then*

$$\Phi: N^1(X) \longrightarrow N_1(X)^\vee, \quad [\omega] \longmapsto ([T] \mapsto T(\omega))$$

is an isomorphism. In particular $N_1(X)$ is finite-dimensional, and every linear functional $\lambda: N_1(X) \rightarrow \mathbb{R}$ has the form $\lambda([T]) = T(\omega)$ for a real closed $(1, 1)$ -form ω with local potentials.

Proof. Injectivity of Φ is the definition of $N^1(X)$: if $T(\omega) = 0$ for every closed current of bidimension $(1, 1)$ then $[\omega] = 0$. For surjectivity, $N^1(X)$ is finite-dimensional by [Proposition 1.7](#), so it suffices to see $\dim N_1(X) \leq \dim N^1(X)$; and the map $\psi: N_1(X) \rightarrow N^1(X)^*$, $[T] \mapsto ([\omega] \mapsto T(\omega))$, is injective by the definition of $N_1(X)$. For the last statement, pass to a resolution $\pi: \hat{X} \rightarrow X$: [Theorem 1.9](#) says $N^1(X) \rightarrow N^1(\hat{X})$ is injective, and dualising turns this into surjectivity of $\pi_*: N_1(\hat{X}) \rightarrow N_1(X)$. ■

Curves land in $N_1(X)$ because the current of integration over a curve is closed and positive: for an irreducible curve $C \subset X$ we write $[C]: H_{\text{BC}}^{1,1}(X) \ni \eta \mapsto \int_C \eta$.

Warning. Every notion below — nef in particular — is defined only for **compact** X . Without properness the definitions say nothing.

1.5 The dictionary

The whole subject is organised by a single substitution.

projective	Kähler
ample divisor	Kähler class
effective divisor	closed positive current
big divisor (sections grow)	class containing a Kähler current
pseudoeffective divisor	class containing a positive current
$\overline{\text{NE}}(X)$, curves	$\overline{\text{NA}}(X)$, positive currents of bidimension $(1, 1)$
linear system, base locus	Lelong numbers, non-Kähler locus

2 Positivities: Kähler, big, pseudoeffective, nef, semi-ample

The four tiers of positivity, treated as objects: how each is defined, what each is stable under, and how they sit inside one another. The organising principle is stated once here and then used repeatedly.

Kähler and big are open conditions; nef and pseudoeffective are closed ones.

Open conditions survive perturbation and are what one carries along inside a construction; closed conditions survive passage to the limit and are what one concludes at the end of one. Each pair is related by taking interiors and closures (Corollary 2.6, Proposition 2.25), and each tier absorbs the weaker one on its own side (Lemma 2.5, Proposition 2.25). The fifth tier, semi-ampleness, is different in kind: it is what turns a class into a morphism, and it is treated last.

2.1 Kähler classes

Definition 2.1 ([HP16, Definition 2.2]). An analytic variety X is **Kähler** if there is a positive closed real $(1, 1)$ -form $\omega \in \mathcal{A}_{\mathbb{R}}^{1,1}(X)$ such that for every $x \in X_{\text{sing}}$ there are an open neighbourhood $x \in U \subseteq X$, a closed embedding $\iota_U: U \hookrightarrow V \subseteq \mathbb{C}^N$, and a smooth **strictly plurisubharmonic** $f: V \rightarrow \mathbb{R}$ with $\omega|_{U \cap X_{\text{sm}}} = (i\partial\bar{\partial}f)|_{U \cap X_{\text{sm}}}$. The **Kähler cone** $\mathcal{K}_X \subset H_{\text{BC}}^{1,1}(X)$ is the cone generated by the classes of Kähler forms.

Intuition. Two features of the definition, rather than the definition itself, are what get used. First, a Kähler form is locally dd^c of a function, so it is a form with local potentials in the sense of Definition 1.4 and therefore has a Bott–Chern class: this is why the theory can be run on $H_{\text{BC}}^{1,1}$ and not on some larger group. Second, the potential is *strictly* plurisubharmonic. Every weaker tier below is obtained by degrading exactly this word — psh instead of strictly psh, or a lower bound $-\epsilon\omega$ instead of ≥ 0 .

Observation 2.2. A Kähler class is nonzero.

Proof. A global potential φ with $\omega = dd^c\varphi$ would be a psh function on a compact space, hence constant, forcing $\omega = 0$. ■

2.1.1 The Kähler cone is open, convex, and contains no line

Proposition 2.3. Let X be a compact Kähler manifold. Then $\mathcal{K}_X \subset H_{\text{BC}}^{1,1}(X)$ is a non-empty open convex cone containing no straight line.

Proof. Convexity and the cone property are immediate: a sum of two Kähler forms is Kähler, and a positive multiple of a Kähler form is Kähler. Salience follows: if α and $-\alpha$ were both Kähler then $0 = \alpha + (-\alpha)$ would be Kähler, contradicting Observation 2.2. Openness is Lemma 2.5 below, which gives $\alpha + \omega \in \mathcal{K}_X$ for every $\omega \in \mathcal{K}_X$ and every α in a neighbourhood of 0 (such an α is a difference of nef classes). ■

Remark 2.4. Saliency is what makes extremal-ray arguments possible at all: a salient closed convex cone is the convex hull of its extremal rays. The same property holds for $\text{Nef}(X)$, $\mathcal{E}_X$ and $\overline{\text{NA}}(X)$.

2.1.2 Kähler + nef = Kähler

This one lemma carries a large part of the formal structure of the subject. Its statement uses the definition of nefness from [Definition 2.39](#); nothing else in [Section 2.4](#) is needed.

Lemma 2.5. *If ω is a Kähler class and α is nef, then $\alpha + \omega$ is Kähler. The same holds if α is merely closed semi-positive with local potentials.*

Proof. By [Definition 2.39](#) there is, for each $\epsilon > 0$, a smooth f_ϵ with $\alpha + i\partial\bar{\partial}f_\epsilon \geq -\epsilon\omega$. Taking $\epsilon < 1$,

$$\alpha + \omega + i\partial\bar{\partial}f_\epsilon \geq (1 - \epsilon)\omega > 0,$$

which is a Kähler form in the class $\alpha + \omega$. If α is semi-positive, take $f_\epsilon = 0$. ■

Corollary 2.6. $\text{Nef}(X) = \overline{\mathcal{K}_X}$ and $\mathcal{K}_X = \text{int Nef}(X)$.

Remark 2.7. Every “perturb and stay positive” step in the Kähler MMP is [Lemma 2.5](#). It is why a nef class is a limit of Kähler classes, why one may add a small Kähler class to restore strict positivity before applying a criterion, and why perturbing by a very general Kähler class to isolate a single extremal ray is legitimate. The semi-positive case is what is used when α is a pull-back $f^*\omega_Y$ ([Section 2.5](#)).

2.1.3 Strict positivity on subvarieties

Lemma 2.8 (Barth–Hulek–Peters–Van de Ven, [Lemma 13.1](#)). *Let X be a compact Kähler manifold, ω a Kähler class and $V \subset X$ an irreducible analytic subvariety with $p = \dim V > 0$. Then*

$$\int_V \omega^p > 0.$$

Proof. The restriction $\omega|_{V_{\text{reg}}}$ is a Kähler form on V_{reg} , so $\omega^p|_{V_{\text{reg}}}$ is a positive volume form and its integral is positive. ■

Corollary 2.9. *A Kähler class is nef and big, and is strictly positive on $\overline{\text{NA}}(X) \setminus \{0\}$. In particular $\mathcal{K}_X \subseteq \mathcal{B}_X \cap \text{Nef}(X)$.*

Proof. Nefness is immediate from [Definition 2.39](#) with $f_\epsilon = 0$. Taking $V = X$ in [Lemma 2.8](#) gives $\int_X \omega^n > 0$, hence $v(\omega) > 0$ and bigness by [Theorem 2.23](#); alternatively ω is itself a Kähler current. Positivity on $\overline{\text{NA}}(X)$ is [Lemma 2.8](#) applied to the generators $\int_V (-) \wedge \omega^{d-1}$ of that cone ([Section 3](#)). ■

Remark 2.10. The conditions in [Lemma 2.8](#) are *necessary*. How far they are from sufficient is the entire content of the Demailly–Păun theory ([Section 4](#)), and the answer is: sufficient, up to passing to the correct connected component.

2.1.4 Pull-back, push-forward and restriction of Kähler classes

The pattern is that Kählerness restricts well, pulls back badly, and pushes forward only as a current.

Lemma 2.11 (Restriction). *If $\iota: V \hookrightarrow X$ is a subvariety and ω is Kähler on X , then $\iota^*\omega$ is Kähler on V .*

Proof. Positivity and closedness of a form are preserved by pull-back along an immersion. For the local potentials: around a point of V , compose the local embedding $U \hookrightarrow \Omega \subset \mathbb{C}^N$ of [Definition 2.1](#) with $V \cap U \hookrightarrow U$, so the same strictly psh f on Ω serves as a local potential for $\iota^*\omega$. A strictly psh function restricted to a complex submanifold of Ω is again strictly psh, since the complex Hessian restricts to the complex Hessian. ■

Fact 2.12 (Pull-back). For $f: X \rightarrow Y$ bimeromorphic or a fibre space, $f^*\omega_Y$ is only *semi-positive*: the complex Hessian of $f^*\varphi$ degenerates in the directions tangent to the fibres of f , exactly where df drops rank. So $f^*\omega_Y$ is a closed semi-positive form with local potentials, and it is Kähler only if f is finite and unramified.

Lemma 2.13 ([DH24]). *Let $f: X \rightarrow Y$ be a birational morphism, α_Y a Kähler form on Y and α_X a relative Kähler form. Then $s f^*\alpha_Y + \alpha_X$ is Kähler for $s \gg 0$.*

Remark 2.14. [Lemma 2.13](#) is the transcendental replacement for “ample + relatively ample = ample”, and it is the workhorse for propagating the Kähler condition down one step of the MMP. In the $\mathbb{Q}$ -factorial case one takes $\alpha_X = -E$ for a relatively ample exceptional divisor E and concludes that $s f^*\alpha_Y - E$ is Kähler.

Fact 2.15 (Blow-ups). If $\pi: \hat{X} \rightarrow X$ is the blow-up of a compact Kähler manifold along a smooth centre with exceptional divisor E , then $\pi^*\omega - \epsilon E$ is Kähler for $0 < \epsilon \ll 1$. In particular the blow-up of a Kähler manifold is Kähler.

The same “subtract a small exceptional class” device reappears in the proof of the Kleiman-type criterion ([Theorem 4.4](#)).

Fact 2.16 (Push-forward). For π bimeromorphic, $\pi_*\omega$ is in general only a Kähler current ([Definition 2.20](#)). This is precisely the mechanism that manufactures Kähler currents in [Section 4](#). Under a rational-singularity hypothesis one can sometimes recover an honest smooth form, by taking a regularised maximum of the local potential with a smooth strictly psh function [[HP16](#), Section 3].

2.1.5 The projectivity criterion

Openness of the Kähler cone ([Proposition 2.3](#)) has one consequence worth isolating, because it is the only bridge in these notes between the transcendental world and the projective one. Recall that

$$\mathrm{NS}(X)_{\mathbb{R}} := (H^{1,1}(X, \mathbb{R}) \cap H^2(X, \mathbb{Q})) \otimes_{\mathbb{Q}} \mathbb{R}$$

is the real Néron–Severi space, i.e. the classes of $\mathbb{R}$ -divisors.

Theorem 2.17 (Projectivity criterion, $\mathbb{R}$ -divisor form). *Let X be a compact Kähler manifold. The following are equivalent.*

1. X is projective.
2. $\mathcal{K}_X \cap H^2(X, \mathbb{Z}) \neq \emptyset$: some Kähler class is integral.
3. $\mathcal{K}_X \cap \mathrm{NS}(X)_{\mathbb{R}} \neq \emptyset$: some Kähler class is the class of an $\mathbb{R}$ -divisor.

Moreover, if X is projective then $\mathcal{K}_X \cap \mathrm{NS}(X)_{\mathbb{R}}$ is exactly the ample cone of X .

Proof. (2) $\Rightarrow$ (1) is the Kodaira embedding theorem, and (1) $\Rightarrow$ (2) is the class of an ample divisor. (2) $\Rightarrow$ (3) is trivial, since an integral class lies in $\mathrm{NS}(X)_{\mathbb{R}}$.

For (3) $\Rightarrow$ (2), let $\alpha \in \mathcal{K}_X \cap \mathrm{NS}(X)_{\mathbb{R}}$. By [Proposition 2.3](#) the cone $\mathcal{K}_X$ is open in $H^{1,1}(X, \mathbb{R})$, so $\mathcal{K}_X \cap \mathrm{NS}(X)_{\mathbb{R}}$ is a non-empty open subset of $\mathrm{NS}(X)_{\mathbb{R}}$. Rational points are dense in $\mathrm{NS}(X)_{\mathbb{R}}$ by construction, so this open set contains a class β with $\beta \in H^2(X, \mathbb{Q})$. Clearing denominators, $m\beta$ is an integral Kähler class for suitable $m \in \mathbb{Z}_{>0}$, since $\mathcal{K}_X$ is a cone.

For the last statement: an ample class is Kähler and lies in $\mathrm{NS}(X)_{\mathbb{R}}$; conversely a Kähler class in $\mathrm{NS}(X)_{\mathbb{R}}$ is a positive real combination of integral Kähler classes by the density argument just given, hence a positive combination of ample classes, hence ample. ■

Remark 2.18. It is the *openness* of $\mathcal{K}_X$ that does the work: it converts a real condition into a rational one for free. This is the first of several places where “Kähler is an open condition” is not a technical convenience but the whole argument. It also explains the shape of the failure: a general compact Kähler manifold has $\mathrm{NS}(X)_{\mathbb{R}}$ too small — possibly zero — for the intersection to be non-empty, and no amount of positivity on X will fix that.

Example 2.19 (K3 surfaces). On a K3 surface the Kähler cone is cut out inside the positive cone of the hyperbolic lattice $H^{1,1}(X, \mathbb{R})$ by the conditions $\omega \cdot C > 0$ over the (-2) -curves, and the ample cone is its intersection with $\text{NS}(X)$, in accordance with [Theorem 2.17](#). This is the cleanest place to see [Proposition 2.3](#), [Lemma 2.8](#) and [Theorem 2.17](#) acting together.

2.2 Kähler currents and big classes

Definition 2.20 ([\[DH24, Definition 2.2\]](#)). Let X be a normal compact analytic variety. A closed positive $(1, 1)$ -current T on X with local potentials is a **Kähler current** if $T \geq \omega$ for some smooth Hermitian form ω on X . A class $\alpha \in H_{\text{BC}}^{1,1}(X)$ is **big** if it contains a Kähler current.

 **Intuition.** Bigness is defined *transcendentally*, through curvature: no global sections are involved, so the notion makes sense when there is no linear system at all. A big class is exactly a *degenerate Kähler class* — the Kähler form is allowed to acquire singularities, and the whole difficulty of the theory is controlling where.

2.2.1 Four descriptions of bigness

Theorem 2.21. Let X be compact Kähler with a fixed Kähler form ω and $\alpha \in H_{\text{BC}}^{1,1}(X)$. The following are equivalent.

1. α contains a Kähler current.
2. $\alpha - \epsilon\{\omega\}$ is pseudoeffective for some $\epsilon > 0$.
3. α contains a Kähler current with analytic singularities.
4. $v(\alpha) > 0$.

Proof. (1) $\iff$ (2) is the definition read twice: a current $T \in \alpha$ with $T \geq \epsilon\omega$ is the same thing as a positive current $T - \epsilon\omega \in \alpha - \epsilon\{\omega\}$. (1) $\implies$ (3) is Demailly regularization [\[Dem92\]](#), and (3) $\implies$ (1) is trivial. (1) $\iff$ (4) is [Theorem 2.23](#). ■

Remark 2.22. Implication (1) $\implies$ (3) is the one that is actually used. A Kähler current with analytic singularities is a genuine Kähler metric on a Zariski open subset, so bigness becomes a statement one can do geometry with. This is the precise sense in which “big + nef $\simeq$ singular Kähler metric”.

Theorem 2.23 (Grauert–Riemenschneider criterion; [Bou02, Theorem 4.8]). *A pseudoeffective class α on a compact Kähler manifold is big if and only if $v(\alpha) > 0$.*

💡 Idea of proof. One direction is immediate: if $T \geq \omega$ is a Kähler current in α then $T_{ac} \geq \omega$, because $T_{ac} - \omega = (T - \omega)_{ac}$ is the absolutely continuous part of a positive current; hence $\int_X T_{ac}^n \geq \int_X \omega^n > 0$. The converse is the substantial half: take $S_k \in \alpha$ with $\int_X S_{k,ac}^n \rightarrow v(\alpha)$, regularise to currents T_k with analytic singularities and $T_k \geq -\varepsilon_k \omega$, resolve $\mu_k^\star T_k = \theta_k + E_k$, and apply the nef-and-big criterion to $\alpha_k := \{\theta_k + \varepsilon_k \mu^\star \omega\}$ to produce a Kähler current in $\alpha + \varepsilon_k \{\omega\}$. The whole point of the argument is that the resulting lower bound $\delta > 0$ **can be taken uniform in k** ; this is where the estimates must be done by hand.

Remark 2.24. Theorem 2.23 is the transcendental replacement for “ $h^0(mL)$ grows like m^n ”. It says the curvature-side and growth-side definitions of bigness still agree when there are no sections to count.

2.2.2 The big cone is open, and absorbs pseudoeffective classes

Proposition 2.25. *Let $\mathcal{B}_X$ be the big cone and $\mathcal{E}_X$ the pseudoeffective cone. Then*

1. $\mathcal{B}_X + \mathcal{E}_X \subseteq \mathcal{B}_X$;
2. $\mathcal{B}_X = \text{int}(\mathcal{E}_X)$ and $\mathcal{E}_X = \overline{\mathcal{B}_X}$.

Proof. (1) If $T \geq \varepsilon \omega$ is a Kähler current in α and $S \geq 0$ is a positive current in β , then $T + S \geq \varepsilon \omega$ is a Kähler current in $\alpha + \beta$. (2) By Theorem 2.21(2), α is big iff $\alpha - \varepsilon \{\omega\} \in \mathcal{E}_X$ for some $\varepsilon > 0$, which says exactly that α is interior to $\mathcal{E}_X$; the second equality follows since $\mathcal{E}_X$ is closed (Proposition 2.32). ■

Since $\mathcal{K}_X \subset \mathcal{B}_X$, part (1) contains “big + Kähler = big” and “big + nef = big” as special cases. This is the exact analogue of Lemma 2.5 one tier down, and is used the same way: to perturb without leaving the cone.

2.2.3 Pull-back and push-forward of big classes

Proposition 2.26. *Let $\pi: \hat{X} \rightarrow X$ be a proper bimeromorphic morphism of compact Kähler varieties and $\alpha \in H_{\text{BC}}^{1,1}(X)$. Then*

$$\alpha \text{ is big} \iff \pi^* \alpha \text{ is big},$$

and if $\hat{\alpha} \in H_{\text{BC}}^{1,1}(\hat{X})$ is big then $\pi_ \hat{\alpha}$ is big.*

Proof. ($\Rightarrow$) Let $T \in \alpha$ be a Kähler current, $T \geq \epsilon \omega_X$. Then $\pi^* T \in \pi^* \alpha$ is a closed positive current with local potentials (pull back the potentials) and $\pi^* T \geq \epsilon \pi^* \omega_X$. Now $\pi^* \{\omega_X\}$ is nef by Proposition 2.48 and big, since $\int_{\hat{X}} (\pi^* \omega_X)^n = \int_X \omega_X^n > 0$ and Theorem 2.23 applies. So $\pi^* \{\omega_X\}$ contains a Kähler current $S \geq \delta \omega_{\hat{X}}$, and $\pi^* T + \epsilon(S - \pi^* \omega_X)$ is a Kähler current in $\pi^* \alpha$.

($\Leftarrow$) Let $\hat{T} \in \pi^* \alpha$ satisfy $\hat{T} \geq \epsilon \pi^* \omega_X$. Push forward: π_* preserves positivity of currents, so $\pi_* \hat{T} - \epsilon \pi_* \pi^* \omega_X \geq 0$. Since π is bimeromorphic, $\pi_* \pi^* \omega_X = \omega_X$, so $\pi_* \hat{T} \geq \epsilon \omega_X$ is a Kähler current in α . The same computation with $\pi^* \omega_X$ replaced by $\omega_{\hat{X}}$ and $\pi_* \omega_{\hat{X}}$ a Kähler current downstairs gives the statement for a general big $\hat{\alpha}$. ■

Remark 2.27. This two-sided invariance is what makes bigness usable along an MMP: bigness of a class can be tested on any bimeromorphic model. It is the opposite of the behaviour of nefness (Proposition 2.48), which pulls back but does not push forward. A relative-to-absolute statement in the same family: a relatively big class plus a large multiple of the pull-back of a big class from the base is big — the bigness counterpart of Lemma 2.13.

2.2.4 Restriction of big and nef classes

Restriction behaves quite differently for the two notions, and the difference is worth stating plainly, because it is a common source of error.

Proposition 2.28. *Let $\iota: V \hookrightarrow X$ be an irreducible subvariety and $\alpha \in H_{\text{BC}}^{1,1}(X)$ nef. Then $\alpha|_V$ is nef.*

Proof. Fix a Kähler form ω on X . By Proposition 2.40(1), $\alpha + \epsilon\{\omega\}$ is Kähler for every $\epsilon > 0$; by Lemma 2.11 its restriction $(\alpha + \epsilon\{\omega\})|_V$ is Kähler on V . That is, $\alpha|_V + \epsilon\{\omega|_V\}$ is Kähler for every $\epsilon > 0$, which by Proposition 2.40(1) applied on V says exactly that $\alpha|_V$ is nef. ■

⚠ Warning. **Bigness does not restrict.** If α is big and $V \subset X$ is a subvariety, $\alpha|_V$ need not be big: a Kähler current $T \in \alpha$ may degenerate along V , and restricting it there destroys the strict lower bound $T \geq \epsilon \omega$. The correct statement quantifies over the locus where the current degenerates: if V is not contained in the locus where every

Kähler current in α has positive Lelong numbers — the transcendental analogue of the augmented base locus — then $\alpha|_V$ is big. This is why, in practice, one restricts *nef and big* classes to *general* members of a family rather than to an arbitrary subvariety.

Lemma 2.29 ([DH25, Lemma 5.3]). *Let X be a normal compact Kähler variety and $f: X \rightarrow C$ a proper surjective morphism to a smooth projective curve. If $\omega \in H_{\text{BC}}^{1,1}(X)$ is nef and big, then $\omega|_F$ is nef and big for a general fibre F of f .*

The nef half of Lemma 2.29 is Proposition 2.28 and holds for *every* fibre; it is the big half that needs the fibre to be general.

Fact 2.30. A nef and big class decomposes as a modified Kähler class plus an effective divisor.

2.3 Pseudoeffective classes

Definition 2.31. A class $\alpha \in H_{\text{BC}}^{1,1}(X)$ is **pseudoeffective** if it contains a closed positive $(1,1)$ -current. The pseudoeffective cone is denoted $\mathcal{E}_X$.

The two elementary comparisons: α is pseudoeffective iff $\alpha + \epsilon\omega$ is big for every $\epsilon > 0$ (Proposition 2.25), and nef $\Rightarrow$ pseudoeffective (Proposition 2.40).

2.3.1 The pseudoeffective cone is closed

Proposition 2.32. $\mathcal{E}_X \subset H_{\text{BC}}^{1,1}(X)$ is a closed convex cone containing no line.

Proof. Convexity is immediate: a sum of positive currents is positive. For closedness, let $\alpha_k \rightarrow \alpha$ with $\alpha_k \in \mathcal{E}_X$, and choose positive currents $T_k \in \alpha_k$. The mass $\int_X T_k \wedge \omega^{n-1}$ is a cohomological quantity, computed from α_k alone, hence bounded along a convergent sequence of classes. Bounded sets of positive currents are weakly compact, so after passing to a subsequence $T_k \rightarrow T$ weakly, with $T \geq 0$ and T closed. Since the class map is continuous for the weak topology, $T \in \alpha$, so $\alpha \in \mathcal{E}_X$. Salience: if $\alpha, -\alpha \in \mathcal{E}_X$ then α is represented by a positive current T and $-\alpha$ by a positive current S , so $T + S$ is a positive current in the class 0; pairing with ω^{n-1} gives $T + S = 0$, hence $T = S = 0$ and $\alpha = 0$. ■

⚠ Warning. $\mathcal{E}_X$ is closed, but it is *not* the closure of a cone of effective divisors — on a general Kähler variety there may be no effective divisors at all. It is defined by currents from the start. This is the same phenomenon that forced the introduction of $H_{\text{BC}}^{1,1}$ in

Section 1: on the divisor side there is nothing to take a closure of.

2.3.2 Siu decomposition

Theorem 2.33 (Siu). *Every closed positive $(1, 1)$ -current T decomposes as*

$$T = \sum_j v(T, D_j) [D_j] + R,$$

where the D_j are the prime divisors with $v(T, D_j) > 0$ (a countable, locally finite family) and R is a closed positive current all of whose divisorial Lelong numbers vanish.

Remark 2.34. Theorem 2.33 is the structural statement that makes the whole “modified” tier possible. Applied to a current with minimal singularities in α , it produces the divisorial Zariski decomposition $\alpha = P(\alpha) + N(\alpha)$ with $P(\alpha)$ modified nef and $N(\alpha)$ effective; the residual part R is what modified nefness is designed to see. It is also the source of the fact that $\partial\mathcal{E}_X$ is locally polyhedral away from the modified nef cone, with extremal rays generated by exceptional prime divisors.

2.3.3 Pull-back and push-forward of pseudoeffective classes

Proposition 2.35. *Let $\pi: \hat{X} \rightarrow X$ be a proper bimeromorphic morphism of compact Kähler varieties. Then $\alpha \in H_{\text{BC}}^{1,1}(X)$ is pseudoeffective if and only if $\pi^*\alpha$ is, and π_* carries pseudoeffective classes to pseudoeffective classes.*

Proof. If $T \in \alpha$ is closed positive with local potentials, then π^*T is defined by pulling back the potentials, is again closed positive with local potentials, and lies in $\pi^*\alpha$. Conversely, if $\hat{T} \in \pi^*\alpha$ is closed positive, then $\pi_*\hat{T}$ is closed and positive — push-forward of currents preserves both — and its class is $\pi_*\pi^*\alpha = \alpha$, since π is bimeromorphic. The same computation gives the last statement. ■

Remark 2.36. The one point requiring care is that on a singular space the push-forward of a current with local potentials need not have local potentials; the statement above is about classes in $H_{\text{BC}}^{1,1}$, where Proposition 1.5 supplies a representative with local potentials on either side.

An alternative and often more convenient proof of both directions: by Proposition 2.25, α is pseudoeffective iff $\alpha + \epsilon\{\omega\}$ is big for every $\epsilon > 0$, and bigness is a two-sided bimeromorphic invariant by Proposition 2.26.

Fact 2.37. Pseudoeffectivity is preserved under small deformation — in contrast to Kählerness of a singular space.

Remark 2.38. Pseudoeffectivity is the hypothesis in almost every theorem of these notes — Boucksom’s nefness criterion ([Proposition 2.46](#)), the divisorial Zariski decomposition, the BDPP circle ([Section 3.2](#)). The reason is structural: it is exactly the condition under which the class contains a current with minimal singularities, and each of those theorems is read off from that current.

2.4 Nefness

Nefness is the terminal condition of the programme: a minimal model is by definition one on which $K_X + B$ is nef, so every step of the MMP is an attempt to manufacture nefness, and every criterion below is a way of certifying that it has been achieved. It is also the tier where the transcendental theory diverges most sharply from the projective one.

2.4.1 The definition

Definition 2.39 ([[HP16](#), Definition 3.10]; [[P98](#)]). Let X be an irreducible reduced **compact** complex space and $u \in H_{\text{BC}}^{1,1}(X)$ a class represented by a form α with local potentials. Then u is **nef** if for some positive $(1, 1)$ -form ω on X and every $\epsilon > 0$ there is $f_\epsilon \in \mathcal{A}^0(X)$ with

$$\alpha + i\partial\bar{\partial}f_\epsilon \geq -\epsilon \omega.$$

In line bundle terms: L is nef if for every $\epsilon > 0$ there is a smooth Hermitian metric h_ϵ with $\Theta_{h_\epsilon} \geq -\epsilon \omega$.

⚠ Warning. The definition demands a **smooth** f_ϵ . This is precisely why Demailly regularization is needed at almost every step where nefness is verified — see the proof of [Proposition 2.46](#).

Relative version ([[DHY23](#)]): β is relatively nef if $\beta + \omega$ is relatively Kähler for every relatively Kähler ω .

2.4.2 Basic properties

Proposition 2.40. *Let X be compact Kähler with a fixed Kähler form ω .*

1. *A class α is nef if and only if $\alpha + \epsilon\{\omega\}$ is Kähler for every $\epsilon > 0$; equivalently, iff α is a limit of Kähler classes.*
2. *$\text{Nef}(X)$ is a closed convex cone containing no line, and $\text{Nef}(X) + \mathcal{K}_X \subseteq \mathcal{K}_X$.*
3. *$\text{Nef} \Rightarrow \text{pseudoeffective}$.*

Proof. (1) If α is nef then $\alpha + \epsilon\{\omega\}$ is Kähler by [Lemma 2.5](#). Conversely, if $\alpha + \epsilon\{\omega\}$ is Kähler for every $\epsilon > 0$, pick a Kähler form $\alpha + \epsilon\omega + i\partial\bar{\partial}f_\epsilon > 0$ in that class; then $\alpha + i\partial\bar{\partial}f_\epsilon \geq -\epsilon\omega$, which is [Definition 2.39](#). The reformulation as a limit is immediate.
 (2) Convexity and the cone property follow from (1), since the Kähler cone is convex; closedness follows from (1) as well, a limit of limits of Kähler classes being one. Saliency follows from that of $\mathcal{K}_X$ ([Proposition 2.3](#)) by passing to closures, using $\int_X \alpha^n \geq 0$ for nef α . The last inclusion is [Lemma 2.5](#).
 (3) By (1), $\alpha + \epsilon\{\omega\}$ is Kähler, hence pseudoeffective, for all $\epsilon > 0$. Letting $\epsilon \rightarrow 0$ and using that $\mathcal{E}_X$ is closed ([Proposition 2.32](#)) gives $\alpha \in \mathcal{E}_X$. ■

Together with [Proposition 2.25](#) this yields the skeleton of these notes:

$$\mathcal{K}_X \subset \text{Nef}(X) \subset \mathcal{E}_X, \quad \mathcal{K}_X \subset \mathcal{B}_X \subset \mathcal{E}_X, \quad \text{Nef}(X) = \overline{\mathcal{K}_X}, \quad \mathcal{E}_X = \overline{\mathcal{B}_X}.$$

Remark 2.41. Nef is a *closed* condition, Kähler an *open* one, and this is why the two are used for different jobs. Kählerness survives perturbation, so it is what one carries along during a construction; nefness survives passage to the limit, so it is what one concludes at the end of one. The base point free theorem ([Theorem 2.52](#)) is the machine that converts the closed condition back into geometry.

2.4.3 Metric versus numerical nefness

This is the single most important place where projective intuition breaks.

Definition 2.42. α is **metrically nef** if [Definition 2.39](#) holds, and **numerically nef** if $\alpha \cdot C \geq 0$ for every irreducible curve $C \subset X$.

Fact 2.43. On a compact Kähler variety, metrically nef $\Rightarrow$ numerically nef. The converse is **false**.

Example 2.44 ([DH25, p. 8]). Let X be a compact Kähler surface of algebraic dimension one, with elliptic fibration $\varphi: X \rightarrow B$ onto a projective curve B . Take H ample on B and

set $u := -\varphi^*H$, so u is *anti*-nef. Every curve $C \subset X$ is contracted by φ , hence $u \cdot C = 0$ for all C . So u is numerically nef and metrically anti-nef simultaneously.

Remark 2.45. The moral is that on a non-algebraic Kähler space *curves see too little*. This is exactly why $\overline{\text{NE}}(X)$ must be enlarged to $\overline{\text{NA}}(X)$ in [Section 3](#). The two notions do agree on algebraic and Moishezon varieties, and — crucially for the MMP — for $K_X + B$ on a Kähler 3-fold under a pseudoeffectivity hypothesis. That last statement is Step 1 of the Das–Hacon proof of the cone theorem for Kähler 3-fold dlt pairs: the argument needs metric and numerical nefness to coincide *for the class* $K_X + B$, via [Proposition 2.46](#).

2.4.4 Boucksom’s nefness criterion

Write $v(\alpha, x)$ for the minimal multiplicity (generic Lelong number) of α at x .

Proposition 2.46 ([\[Bou04, Proposition 3.2\]](#)). *Let $\alpha \in H_{\text{BC}}^{1,1}(X)$ be pseudoeffective. Then*

1. α is nef $\iff v(\alpha, x) = 0$ for every $x \in X$;
2. α is modified nef $\iff v(\alpha, D) = 0$ for every prime divisor D .

Proof. ($\implies$) If α is nef, then for each $\epsilon > 0$ the smooth form $\theta_\epsilon := \alpha + i\partial\bar{\partial}f_\epsilon$ is an almost positive current in the class α with $\theta_\epsilon \geq -\epsilon\omega$. A smooth form has vanishing Lelong number at every point, so the infimum over ϵ defining $v(\alpha, x)$ is 0.

($\impliedby$) Suppose $v(\alpha, x) = 0$ for all x . For each $\epsilon > 0$ let $T_{\min, \epsilon}$ be a current with minimal singularities in the class $\alpha[-\epsilon\omega]$, i.e. among closed currents $T \in \alpha$ with $T \geq -\epsilon\omega$; such a current exists because α is pseudoeffective. Minimal multiplicity is computed by these currents, so $v(T_{\min, \epsilon}, x) = 0$ for every x . Applying Demailly regularization [\[Dem92\]](#) to $T_{\min, \epsilon}$ replaces it by a *smooth* form in α with lower bound $-\epsilon'\omega$ for any $\epsilon' > \epsilon$, the loss of positivity being controlled by the Lelong numbers, which vanish. Since $\epsilon' > 0$ was arbitrary this is exactly [Definition 2.39](#). ■

Remark 2.47. The criterion converts nefness — a statement about the existence of infinitely many smooth approximating potentials — into a statement about *one number at each point*, computed from the class alone. Every verification of nefness that does not go through the cone ([Theorem 4.4](#)) goes through this. The second clause is what defines the “modified” tier, and its projective shadow is the statement that a pseudoeffective divisor is nef iff its asymptotic vanishing order is 0 along every divisorial valuation.

2.4.5 Pull-back and push-forward of nef classes

Proposition 2.48. *Let $f: X \rightarrow Y$ be a surjective morphism of compact Kähler varieties and $\alpha \in H_{\text{BC}}^{1,1}(Y)$.*

1. *If α is nef then $f^*\alpha$ is nef.*
2. *Conversely, if $f^*\alpha$ is nef then α is nef [P98].*
3. *Restriction to a subvariety preserves nefness (Proposition 2.28).*

The push-forward of a nef class need not be nef.

Proof of (1). Fix positive forms ω_Y on Y and ω_X on X . Given $\epsilon > 0$, Definition 2.39 supplies a smooth f_ϵ on Y with $\alpha + i\partial\bar{\partial}f_\epsilon \geq -\epsilon\omega_Y$. Pulling back, and using that pull-back preserves the order relation between real $(1,1)$ -forms,

$$f^*\alpha + i\partial\bar{\partial}(f_\epsilon \circ f) \geq -\epsilon f^*\omega_Y.$$

Since X is compact and $f^*\omega_Y$ is a semi-positive form, there is a constant $C > 0$, independent of ϵ , with $f^*\omega_Y \leq C\omega_X$. Hence $f^*\alpha + i\partial\bar{\partial}(f_\epsilon \circ f) \geq -\epsilon C\omega_X$. As $\epsilon > 0$ was arbitrary and $f_\epsilon \circ f$ is smooth, this is Definition 2.39 for $f^*\alpha$. ■

Remark 2.49. Direction (2) is genuinely harder and is Păun's theorem [P98]: it is the statement that numerical effectiveness descends along surjections, and it has no one-line proof of the above kind.

This is the mirror image of Proposition 2.26: nefness pulls back well and pushes forward badly, bigness the reverse. Note also the contrast with bigness in Section 2.2.4: nefness restricts to *every* subvariety, bigness only to those not contained in the degeneracy locus. The one positive statement in the push-forward direction is that the negative part of the push-forward of a nef class under a bimeromorphic morphism vanishes — so nefness is not destroyed *divisorially* by an MMP step, even though it is not preserved outright.

⚠ Warning. In the MMP for Kähler generalized pairs one frequently needs a class to be *NQC* (a non-negative combination of nef classes of the appropriate type) rather than merely nef, because the transcendental cone can fail to be spanned by the classes the induction runs on [DHY23].

2.5 Semi-positivity and semi-ampleness

2.5.1 The implication chain

Proposition 2.50. *For a class on a compact Kähler variety,*

$$\text{Kähler} \implies \text{semi-ample} \implies \text{semi-positive} \implies \text{nef} \implies \text{pseudoeffective}.$$

Proof. Semi-ample $\implies$ semi-positive: a semi-ample class is $f^*\omega_Y$ for a Kähler ω_Y downstairs, and the pull-back of a positive form is semi-positive (Fact 2.12). Semi-positive $\implies$ nef: a semi-positive representative satisfies Definition 2.39 with $f_\epsilon = 0$. Nef $\implies$ pseudoeffective is Proposition 2.40(3). Kähler $\implies$ semi-ample is the case $Y = X$, $f = \text{id}$. ■

Warning. Nef $\not\Rightarrow$ semi-positive. This failure has no projective counterpart worth the name, and it is why nefness is a genuinely weaker and harder condition here: a nef class need not contain *any* smooth semi-positive representative, the best available being the family $\alpha + \epsilon\omega$ of Proposition 2.40(1). Nef $\not\Rightarrow$ semi-ample is already false projectively — that is the abundance problem — and transcendently there is not even a linear system in which to pose the question.

Two supplementary facts of the same kind, useful but not load-bearing here: a semi-ample and strictly nef class is Kähler; and a log canonical class numerically equal to the pull-back of a Kähler class is semi-ample.

Remark 2.51. Semi-ampleness is what turns a *class* into a *morphism*. Nefness certifies that the MMP has stopped; semi-ampleness is what lets one contract, take the Iitaka fibration, or descend the class to a base. This is why the transcendental substitute below is stated as the existence of a fibration f with $\alpha = f^*\omega_Y$, and not as any statement about sections.

2.5.2 The base point free theorem

In the transcendental setting there is no linear system, so “semi-ample” cannot mean “ $|mL|$ is base point free”. The correct replacement is the existence of a semi-ample fibration.

Theorem 2.52 ([DHP24, Theorem 5.5]). *Let (X, B) be a compact Kähler 3-fold klt pair and $\alpha \in N^1(X)$ a nef class such that $\alpha - (K_X + B)$ is nef and big. Then there is a proper morphism $f: X \rightarrow Y$ with connected fibres onto a normal compact Kähler variety Y with rational singularities, and a Kähler class $\omega_Y \in N^1(Y)$, such that $\alpha = f^*\omega_Y$. If $\alpha - (K_X + B)$ is Kähler, then f is projective.*

The dlt version is [DHP24, Corollary 5.6]; the $\mathbb{Q}$ -factorial version is in [DH25]; the generalized pair version is [DHY23, Theorem 3.14]. All of these are stated for 3-folds.

The dimension restriction has since been removed. The base point free conjecture in the Kähler setting — equivalently Tosatti’s conjecture — is now a theorem of Hacon–Xie in arbitrary dimension.

Theorem 2.53 (Transcendental base point free theorem, [HX26, Theorem 1.4]). *Let $(X, B + \beta)$ be a compact Kähler gklt pair such that $K_X + B + \beta_X$ is nef and $B + \beta_X$ is modified big. Then $K_X + B + \beta_X$ is semiample: there is a Moishezon contraction $f : X \rightarrow Y$ onto a compact Kähler space and a Kähler class ω_Y on Y with*

$$K_X + B + \beta_X \equiv f^* \omega_Y.$$

If X is strongly $\mathbb{Q}$ -factorial, then f is projective.

💡 Idea of proof. The argument splits on whether $K_X + B + \beta_X$ is big. In the non-big case one uses Ou’s theorem (Section 3.2) to produce an MRC fibration, runs a relative MMP to a Mori fibre space, and descends by the canonical bundle formula, concluding by induction on the dimension of the base. In the big case one uses that the non-nef locus coincides with the null locus, hence is analytic, places a Kähler current with weak analytic singularities along it, and runs an induction over the nested multiplier-ideal subspaces; the resulting morphisms are glued on a formal neighbourhood of $E_+(\alpha)$ and then algebraized to a global morphism. The proof is the subject of a separate note in this series and is not reproduced here.

Remark 2.54. The statement is about a *class*, not a line bundle, and there is no linear system anywhere in it: “semiample” means $\equiv f^* \omega_Y$ for a morphism f and a Kähler class ω_Y . That reformulation, forced by the absence of divisors, is the whole reason the transcendental statement is harder than its projective counterpart.

3 Mori cone, generalized Mori cone, and duality of cones

Everything so far has been about cones in $N^1(X)$. This section introduces the cones in $N_1(X)$ that they are tested against, and the duality between the two sides — which turns out to be the same statement as the Kählerness criterion of the appendix.

3.1 The two Mori cones

- Definition 3.1** ([HP16, p. 226]; [DH25, Definition 2.2]).
1. $\overline{NA}(X) \subset N_1(X)$ is the closed cone generated by the classes of *positive closed currents*.
 2. $\overline{NE}(X) \subset \overline{NA}(X)$ is the closure of the cone generated by the currents of integration T_C over irreducible curves $C \subset X$.

Proposition 3.2. *The inclusion $\overline{NE}(X) \subseteq \overline{NA}(X)$ is in general strict, already for projective X .*

Proof. $\overline{NE}(X)$ spans a space of dimension $\rho(X)$, while $\overline{NA}(X)$ spans $H^{n-1,n-1}(X)$. For a projective K3 surface the Torelli theorem gives $\rho(X) \leq 20 = h^{1,1}(X)$, so $\overline{NE}(X)$ lies in a proper subspace $N_1(X) \subsetneq H_{1,1}(X, \mathbb{R})$ while $\overline{NA}(X)$ contains classes outside it. More dramatically, there are K3 surfaces of Picard number 0: no curves at all, yet plenty of positive closed currents. ■

Remark 3.3. This is the cone-side shadow of [Example 2.44](#): curves are not enough to detect positivity, and $\overline{NA}(X)$ is the smallest enlargement that fixes it. Extremality is not inherited between the two cones, so one must be careful which cone an extremal ray argument is run in.

Fact 3.4 (Generators). $\overline{NA}(X)$ is generated by the functionals $\int_V (-) \wedge \omega^{d-1}$ over subvarieties V . This is what makes [Theorem 4.1](#) applicable to it, and is the reason cone theorem proofs need only check generators of the shape $[\hat{\omega}]^2, [\hat{\omega}] \cdot [\hat{S}], [\hat{C}]$.

$\overline{NA}(X)$ contains no line, which is what makes extremal-ray arguments possible; the push-forward of $\overline{NA}$ under a bimeromorphic morphism is again $\overline{NA}$; and contracting a negative extremal ray of $\overline{NA}$ preserves the Kähler condition. The K_X -negative structure of $\overline{NA}(X)$ — the cone theorem itself — is the subject of a separate note in this series and is only invoked here.

3.2 Duality of cones

3.2.1 Nef(X) and $\overline{NA}(X)$

Theorem 3.5 ([HP16, Proposition 3.15]; [DHP24, Proposition 2.4]; [DP04, Corollary 0.4]). *Let X be a normal compact Kähler variety with **rational singularities**. Then $\text{Nef}(X)$ and $\overline{\text{NA}}(X)$ are dual to each other under the isomorphism $\Phi: N^1(X) \rightarrow N_1(X)^*$ of Proposition 1.13. In particular $\omega \in H_{\text{BC}}^{1,1}(X)$ is Kähler if and only if $\omega \cdot \gamma > 0$ for every $\gamma \in \overline{\text{NA}}(X) \setminus \{0\}$.*

The second sentence is exactly the Kleiman-type criterion Theorem 4.4: duality and the Kählerness criterion are two faces of one statement. The easy half — that a Kähler class is strictly positive on $\overline{\text{NA}}(X) \setminus \{0\}$ — is Corollary 2.9. The projective baseline for comparison is $\text{Nef}(X)^* = \overline{\text{NE}}(X)$; the manifold version, $\mathcal{N}^* = \overline{\mathcal{K}}$ together with $\text{Eff}^*(X) = \text{Nef}(X)$, is [Dem12, 19.6].

3.2.2 Pseudoeffective versus movable: BDPP

Theorem 3.6 ([BDPP13], projective case). *On a projective manifold the pseudoeffective cone $\mathcal{E}$ is dual to the movable cone $\mathcal{M}$: a class is pseudoeffective iff it has non-negative intersection with every curve moving in a family covering X .*

⚠ Warning. Two statements are both called “BDPP” and must be kept apart.

- (a) *Cone duality.* [BDPP13, Conjecture 2.3]: for an arbitrary compact Kähler manifold, $\mathcal{E}$ and $\mathcal{M}$ are dual. This is **still open** in the Kähler case.
- (b) *Uniruledness.* K_X is pseudoeffective iff X is not uniruled. For compact Kähler manifolds this is now a theorem of Ou [Ou25].

It is (b) that Hacon–Păun assume in order to deduce the cone theorem in arbitrary dimension. Conflating (a) with (b) misstates what is known.

Statement (b) is proved in full in a separate note of this series, following Ou [Ou25]; the projective case of the duality statement, following [BDPP13], is treated there as well. We use both freely below.

Other dualities on the same circle: the cone of nef curves is dual to the pseudoeffective cone; Serre duality induces a pairing between the transcendental pseudoeffective and mobile cones, and between the nef cone and the cone of effective $(n-1, n-1)$ -classes.

4 Appendix: Kählerness criterion

Three criteria of increasing usability inside the MMP: an intersection-theoretic one, a topological refinement, and a duality one. All are stated without proof; the proof ideas are recorded because they are what gets reused.

4.1 The Demailly–Păun Nakai–Moishezon criterion

Theorem 4.1 ([DP04, Corollary 2.39]; singular version in [DHP24]). *Let X be a compact normal Kähler variety, ω a Kähler form and $\alpha \in H_{\text{BC}}^{1,1}(X)$. Then*

1. α is nef $\iff \int_V \alpha^k \wedge \omega^{p-k} \geq 0$ for every analytic p -dimensional $V \subset X$ and all $0 < k \leq p$;
2. α is Kähler $\iff$ the same integrals are all > 0 .

There is a variant tailored to MMP applications: starting from a smooth ∂ - and $\bar{\partial}$ -closed $(1,1)$ -form α with $\{\alpha\}$ nef and big and $\int_{V_{\text{reg}}} \alpha^{\dim V} > 0$ for every positive-dimensional reduced analytic V , one concludes that α is Kähler [DHP24, Theorem 2.29].

$$\text{big} + \text{nef} \simeq \text{singular Kähler metric (Kähler current)}.$$

By Demailly regularization the Kähler current can be taken with analytic singularities, i.e. it is an honest smooth Kähler metric on a Zariski open subset.

4.2 The Kähler cone is a connected component of the positive cone

Theorem 4.2 ([DP04, Main Theorem 0.1]; [Dem12, 18.5]). *Let X be a compact Kähler manifold and*

$$\mathcal{P} := \left\{ [\alpha] \in H^{1,1}(X, \mathbb{R}) \mid \int_Y \alpha^{\dim Y} > 0 \text{ for all irreducible analytic } Y \subset X, \dim Y > 0 \right\}.$$

Then $\mathcal{K}$ is one of the connected components of $\mathcal{P}$. If X is projective, $\mathcal{K} = \mathcal{P}$.

Example 4.3 (Generic torus: why “one of the components”). Let $X = \mathbb{C}^n / \Lambda$ be a generic complex torus, which has no analytic subsets other than points and X . Numerical positivity then reduces to the single condition $\int_X \alpha^n > 0$, and $(1,1)$ -classes correspond to constant Hermitian forms on $\mathbb{C}^n$. So $\mathcal{P} = \{\det(\alpha) > 0\}$, and the theorem becomes the linear algebra statement that the positive definite forms are *one* connected component — the others being forms of signature (p, q) with $p + q = n$ and q even. These are not even convex when $p, q > 0$.

4.3 The Kleiman-type criterion

Theorem 4.4 ([HP16, Theorems 3.17–3.18]). *Let X be a normal compact Kähler 3-fold with rational singularities and $[\eta] \in N^1(X)$. Then $[\eta]$ is Kähler if and only if $[\eta]$ is strictly positive on $\overline{NA}(X) \setminus \{0\}$.*

💡 Idea of proof. Take a desingularization $\pi: \hat{X} \rightarrow X$ which is projective with SNC exceptional locus, and an effective π -exceptional D with $-D$ π -ample. For $N \gg 0$ the class $[N\pi^*\eta - D]$ is strictly positive on $\overline{NA}(\hat{X})$, so by Theorem 4.1 it is a Kähler class $[\hat{\omega}]$. Pushing forward, $\pi_*(\hat{\omega}) = N\eta + \partial\bar{\partial}R$ is a Kähler current on X , smooth outside the image E of the exceptional locus. A Kähler current contains a Kähler form iff its restriction to each component Z of the Lelong level sets is Kähler; here dimension three enters: $Z \subseteq E$ forces Z to be a point or a curve, and on a curve the positivity hypothesis $T_Z(\eta) > 0$ says exactly that $i_Z^*\eta$ has positive degree.

⚠ Warning. Two hypotheses cannot be dropped. Dimension 3 is used to force Lelong level sets to be points or curves. Rational singularities is needed for the duality of Theorem 3.5 to hold at all. Whether a Kleiman criterion holds in higher dimension is open.

4.4 Fujiki, Moishezon, projective

Theorem 4.5 ([Dem12, 14.17]; [DP04, Theorem 3.4]; [Bou04, Proposition 2.3]). *A compact complex manifold X is bimeromorphic to a Kähler manifold, i.e. $X \in \mathcal{C}$, if and only if it carries a Kähler current — equivalently, iff the big cone is non-empty.*

💡 Idea of proof. If X is bimeromorphic to a Kähler Y , resolve to a modification $\mu: \tilde{Y} \rightarrow X$ and push forward a Kähler form: compactness gives $\mu^*\omega \leq C\tilde{\omega}$, so $T = \mu_*\tilde{\omega} \geq C^{-1}\omega$ is a Kähler current. Conversely, given a Kähler current T , Demailly regularization replaces it by T' with analytic singularities; on a resolution $\mu^*T' = \lambda[\tilde{D}] + \tilde{\alpha}$ with $\tilde{\alpha}$ smooth and $\tilde{\alpha} \geq \epsilon\mu^*\omega$, and subtracting small multiples of exceptional classes blow-up by blow-up produces an honest Kähler form.

⚠ Warning. Per [DH24, Remark 2.3] it is *not known* whether a singular Fujiki class $\mathcal{C}$ variety carries a big $(1, 1)$ -class. The obstruction is that the Poincaré lemma may fail on a singular space, so the push-forward of a Kähler current need not have local potentials.

Key takeaways

- Positivity lives on $H_{\text{BC}}^{1,1}(X)$, not on divisors; **rational singularities** is what makes $H_{\text{BC}}^{1,1}(X) \hookrightarrow H^2(X, \mathbb{R})$ and hence intersection theory available.
- Kähler and big are *open* conditions; nef and pseudoeffective are *closed* ones. $\text{Nef} = \overline{\mathcal{K}}$, $\mathcal{K} = \text{int Nef}$, $\mathcal{E} = \overline{\mathcal{B}}$, $\mathcal{B} = \text{int } \mathcal{E}$.
- Each tier absorbs the weaker one on its own side: Kähler + nef = Kähler, big + pseudoeffective = big.
- Bigness is a two-sided bimeromorphic invariant; nefness pulls back well but does not push forward.
- Nefness restricts to *every* subvariety; bigness restricts only to those avoiding the locus where the Kähler current degenerates, which is why one restricts to *general* fibres.
- Metric nefness is strictly stronger than numerical nefness on a non-algebraic space — curves see too little — which is why $\overline{\text{NE}}(X)$ must be enlarged to $\overline{\text{NA}}(X)$.
- $\text{Nef}(X)$ and $\overline{\text{NA}}(X)$ are dual, and that duality *is* the Kählerness criterion.

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